

Data Flow Diagram

Project Title: Rising Waters

Introduction

The Data Flow Diagram (DFD) illustrates how data moves through the **Rising Waters** Flood Prediction System. It shows the flow of information from the user to the Machine Learning model and finally to the prediction result.

Input

The user enters weather-related parameters such as temperature, humidity, cloud cover, annual rainfall, seasonal rainfall, average June rainfall, and subdivision rainfall through the web application.

Data Processing

The system validates the entered data and applies feature scaling using the trained scaler. The processed data is then sent to the Machine Learning model for prediction.

Prediction

The trained Machine Learning model analyzes the processed input and predicts whether there is a possibility of flooding. It also calculates the prediction probability.

Output

The prediction result is displayed on the web application as either **Flood** or **No Flood** along with the prediction probability. Based on the result, users can take appropriate preventive actions.

Conclusion

The **Rising Waters** system ensures a smooth and efficient flow of data from user input to prediction output, enabling accurate and timely flood risk analysis through Machine Learning.